CONSEQUENCES OF FINITE GLOBAL DIMENSION FOR TOR DIMENSION AND TOR AMPLITUDE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Criterion for tor amplitude through the cohomology of the complex against modules. Lemma \ref{lemma-cohomology-tor-amplitude}, cited at proof lines 17301-17303. Gives the second and third statements from the first.

Source reference. Original lemma 066P: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7943–7943. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 066I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7937–7937. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring of finite global dimension d. Then*

(1) *every module has tor dimension $\leq d$,*
(2) *a complex of R-modules $K^\bullet$ with $H^i(K^\bullet) \neq 0$ only if $i \in [a, b]$ has tor amplitude in $[a - d, b]$, and*
(3) *a complex of R-modules $K^\bullet$ has finite tor dimension if and only if $K^\bullet \in D^b(R)$.*

Proof. The assumption on R means that every module has a finite projective resolution of length at most d, in particular every module has tor dimension $\leq d$. The second statement follows from Lemma 066I and the definitions. The third statement is a rephrasing of the second. □

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